

SRM VEC
DEPARTMENT OF AIDS
Semester End Examination
(Regulations 2023)
AD3445 – AI1

Branch: B.Tech | Date: 01.02.2026 | Max. Marks: 100 | Duration: 3 Hrs

Answer ALL Questions
(100 Marks)

PART A — (10 × 2 = 20 marks)

1. Define a Rational Agent and list the four factors that depend on the measure of rationality. [CO1] [BTL1-Remember] (2)
2. Differentiate between Breadth-First Search (BFS) and Depth-First Search (DFS) in terms of space complexity. [CO1] [BTL2-Understand] (2)
3. What is a Constraint Satisfaction Problem (CSP)? Give an example. [CO2] [BTL1-Remember] (2)
4. State the purpose of Alpha-Beta pruning in game tree search. [CO2] [BTL2-Understand] (2)
5. Convert the following sentence into First-Order Logic (FOL): 'Every student who takes AI loves logic.' [CO3] [BTL3-Apply] (2)
6. Define Skolemization with a simple example. [CO3] [BTL1-Remember] (2)
7. What are the three components of a STRIPS planning operator? [CO4] [BTL1-Remember] (2)
8. State Bayes' Rule and explain each term. [CO4] [BTL2-Understand] (2)
9. Define a Hidden Markov Model (HMM). [CO5] [BTL1-Remember] (2)
10. What is the difference between Supervised and Unsupervised Learning? [CO5] [BTL2-Understand] (2)

PART B — (5 × 16 = 80 marks)

- 11(a) (i) Explain the A* search algorithm. Prove that A* is optimal if the heuristic function $h(n)$ is admissible. (ii) Compare and contrast Greedy Best-First Search and A* Search.

OR

(b) Explain the various types of Intelligent Agents based on their degree of perceived intelligence and capability. Provide a diagram for a Model-based Reflex Agent.

- 12(a) Apply the Minimax algorithm with Alpha-Beta pruning on a game tree of depth 3 and branching factor 2. Show the pruned branches and explain the pruning condition.

OR

(b) (i) Formulate the 8-Queens problem as a Constraint Satisfaction Problem. (ii) Explain the backtracking search for CSPs and how heuristics like Minimum Remaining Values (MRV) improve it.

- 13(a) (i) Describe the steps involved in converting a First-Order Logic formula into Conjunctive Normal Form (CNF). (ii) Use Resolution to prove that 'Socrates is mortal' given: 'All men are mortal' and 'Socrates is a man'.

OR

(b) Explain the following Knowledge Representation schemes with examples: (i) Semantic Networks (ii) Frames and Scripts

- 14(a) Construct a Bayesian Network for the 'Burglary' problem (Burglary, Earthquake, Alarm, MaryCalls, JohnCalls). Explain how to calculate the joint probability $P(B, \sim E, A, J, M)$ using the network.

OR

(b) (i) Explain the Goal Stack Planning (GSP) algorithm using the Blocks World domain. (ii) Discuss the Sussman Anomaly and why linear planners fail to solve it.

15(a) Explain the ID3 algorithm for Decision Tree Induction. Illustrate with an example how Information Gain and Entropy are used to select the root attribute.

OR

(b) (i) Explain the architecture of an Expert System and its components. (ii) Describe the Q-Learning algorithm in Reinforcement Learning.

Faculty In-Charge

Verified by

HOD